

Business Performance Analysis Using Excel and Power BI

1. Project Overview and Objective

Project Overview:

This project analyzes e-commerce sales data to understand sales performance, customer behavior, and product trends using an interactive dashboard.

Project Objective:

- Analyze overall sales performance.
- Identify top-selling products and categories.
- Track returns and discounts.
- Monitor sales trends.
- Support better business decisions with data insights.

2. Data Sources

- **Source Description and Timeline:** Sample customer data for RFM analysis for 2024-2025.
- **Domain:** E-Commerce

3. Problem Statement

Objectives

- Analyze overall sales performance.
- Identify top-selling products and categories.
- Understand customer purchasing.
- Analyze sales by customer segment and sales channel.
- Evaluate the impact of discounts on sales.
- Monitor returned products.
- Build an interactive dashboard for business decision-making.

Business Questions

1. Which products generate the highest sales?
2. Which product categories perform the best?
3. Who are the top customers by sales?
4. Which customer segment contributes the most revenue?
5. Which sales channel has the highest sales?
6. How do discounts affect total sales?
7. Which products are returned most frequently?
8. What are the monthly sales trends?
9. What is the average order value?
10. Which products should the company focus on to increase?

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Order_Date	Date	Date when the customer places the Order
Order_Id	Whole Number(Integer)	Unique identification number for each order.
Customer_Id	Whole Number(Integer)	Unique identification number for each Customer
Customer	Text(String)Categorical	Customer Name
Segment	Text(string)Categorical	Customer Category
Channel	Text Categorical	Order placed through the Channel
Product_Id	Whole Number	Unique identification number for each product
Product	Whole Number	Name of the product purchased.
Category	Text Categorical	Category Name
Quantity	Decimal Number	<u>No of</u> units ordered by the customer
Unit_Price	Decimal Number	Price per unit
Discount %	Whole Number	Discount percentage allowed in the Product
Line_Total	Decimal Number	Total Price after the discount

Returned	Boolean(0-no,1=yes)	Whether products were returned or not. Using a Boolean expression.
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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Found and removed 4431 duplicate values in the Order_Id column. 5569 values remaining.
- It's improved data accuracy.
- Checked and handled missing values. No missing values in the dataset.
- Standardized date, text, and numeric formats for consistency.
- Removed extra spaces using the TRIM function.
- There is no inconsistent data.
- Verified data types for Order Date, Quantity, Unit Price, Discount %, Created
- **calculated fields**
- **Gross sales=Quantity*Unit Price**
- **Net Sales(Revenue) = Gross Sales – Discount Amount**
- **Discount Amount = Gross Sales × Discount %**
- **Net sale = Net Sales(Revenue)*Returned**

Duplicates in Order id in excel using duplicate tool

1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	order_date	order_id	customer_id	customer	segment	channel	product_id	product	category	quantity	unit_price	discount_pct	returned			
2	2025-08-02	1001	20	Hugo Brooks	Loyal	Paid Search	28	SPF 50 Loti	Beauty	1	23.27	5	0			
3	2025-08-02	1002	41	Owen Nguye	Regular	Organic	5	Socks 3-Pa	Apparel	2	58.34	0	0			
4	2025-08-02	1003	86	Lila Mendez	Regular	Direct	31	Resistance	Sports	1	37.72	25	0			
5	2025-08-02	1004	97	Jamie Romar	Loyal	Organic	5	Socks 3-Pa	Apparel	2	58.34	10	0			
6	2025-08-02	1005	108	Tessa Okafor	Loyal	Referral	32	Foam Roller	Sports	1	51.36	10	0			
7	2025-08-02	1006	113	Cole Okafor	Champion	Referral	29	Yoga Mat	Sports	1	31.47	0	0			
8	2025-08-02	1007	115	Owen Petrov	Champion	Paid Search	1	Crew Tee	Apparel	3	67.83	10	0			
9	2025-08-02	1008	119	Casey Costa	New	Organic	22	Scented Ca	Home	1	33.84	0	0			
10	2025-08-02	1009	138	Priya Sullivan	Regular	Email	40	Art Kit	Toys	1	21.25	0	0			
11	2025-08-02	1010	141	Devon Sullivan	Loyal	Referral	34	Running Be	Sports	1	50.13	0	0			
12	2025-08-02	1011	165	Reese Okafor	Champion	Social	18	LED Lamp	Home	1	54.69	10	0			
13	2025-08-02	1012	343	Reese Costa	Champion	Direct	26	Shampoo B	Beauty	3	29.61	10	0			
14	2025-08-02	1013	379	Mateo Patel	Loyal	Social	31	Resistance	Sports	2	37.72	10	0			
15	2025-08-02	1014	395	Noel Novak	Loyal	Organic	12	Bluetooth S	Electro	1	110.84	10	0			
16	2025-08-02	1015	447	Morgan Ngui	New	Social	19	Cutting Boa	Home	1	11.13	20	0			
17	2025-08-02	1016	470	Jamie Costa	Champion	Paid Search	13	Webcam	Electro	1	60.48	0	0			
18	2025-08-02	1017	568	Hugo Flynn	Loyal	Email	12	Bluetooth S	Electronics	3	110.84	0	0			
19	2025-08-02	1018	593	Reese Carter	Regular	Paid Search	36	Puzzle 500	Toys	1	44.06	0	0			
20	2025-08-02	1019	612	Drew Kowals	New	Email	13	Webcam	Electronics	1	60.48	0	0			
21	2025-08-02	1020	644	Tessa Osei	Loyal	Organic	2	Hoodie	Apparel	1	24.05	0	0			
22	2025-08-02	1021	692	Iris Bauer	Regular	Social	23	Vitamin C S	Beauty	1	20.25	0	0			
23	2025-08-02	1022	772	Maya Fischer	Loyal	Email	25	Sheet Mask	Beauty	1	5.03	25	1			
24	2025-08-02	1023	801	Nora Yamada	Loyal	Organic	10	USB-C Cabl	Electronics	1	29.62	10	0			
25	2025-08-02	1024	823	Maya Park	Regular	Paid Search	34	Running Be	Sports	1	50.13	0	0			
26	2025-08-02	1025	852	Sasha Diaz	Regular	Paid Search	15	Smart Plug	Electronics	3	8.53	0	0			
27	2025-08-02	1026	887	Mateo Noval	Champion	Referral	30	Water Bott	Sports	1	45.18	5	0			
28	2025-08-02	1027	905	Tessa Reid	Loyal	Email	13	Webcam	Electronics	1	60.48	0	0			
29	2025-08-02	1028	906	Noel Flynn	Champion	Social	30	Water Bott	Sports	1	45.18	5	0			
30	2025-08-02	1029	942	Maya Kowals	Loyal	Organic	10	USB-C Cabl	Electronics	1	29.62	0	0			
31	2025-08-02	1030	970	Ezra Sullivan	Regular	Organic	38	Board Gam	Toys	1	15.37	0	0			
32	2025-08-02	1031	1035	Priya Yamada	Loyal	Social	10	USB-C Cabl	Electronics	1	29.62	10	0			
33	2025-08-02	1032	1041	Hugo Reyes	Loyal	Social	16	Throw Pillo	Home	1	10.61	0	0			
34	2025-08-02	1033	1110	Riley Petrov	Loyal	Direct	15	Smart Plug	Electronics	1	8.53	0	0			
35	2025-08-02	1034	1127	Ivy Sullivan	Champion	Organic	9	Wireless Ea	Electronics	1	78.18	10	0			

Microsoft Excel

4431 duplicate values found and removed; 5569 unique values remain. Note that counts may include empty cells, spaces, etc.

OK

	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	product_id	product	category	quantity	unit_price	discount_pct	returned	Cleaned Customer	Cleaned	Cleaned Channel	Cleaned Product	Cleaned Category		
2	28	SPF 50 Lotion	Beauty	1	23.27	5	0	Hugo Brooks	Loyal	Paid Search	SPF 50 Lotion	Beauty		
3	5	Socks 3-Pack	Apparel	2	58.34	0	0	Owen Nguyen	Regular	Organic	Socks 3-Pack	Apparel		
4	31	Resistance Bands	Sports	1	37.72	25	0	Lila Mendez	Regular	Direct	Resistance Bands	Sports		
5	5	Socks 3-Pack	Apparel	2	58.34	10	0	Jamie Romano	Loyal	Organic	Socks 3-Pack	Apparel		
6	32	Foam Roller	Sports	1	51.36	10	0	Tessa Okafor	Loyal	Referral	Foam Roller	Sports		
7	29	Yoga Mat	Sports	1	31.47	0	0	Cole Okafor	Champion	Referral	Yoga Mat	Sports		
8	1	Crew Tee	Apparel	3	67.83	10	0	Owen Petrov	Champion	Paid Search	Crew Tee	Apparel		
9	22	Scented Candle	Home	1	33.84	0	0	Casey Costa	New	Organic	Scented Candle	Home		
10	40	Art Kit	Toys	1	21.25	0	0	Priya Sullivan	Regular	Email	Art Kit	Toys		
11	34	Running Belt	Sports	1	50.13	0	0	Devon Sullivan	Loyal	Referral	Running Belt	Sports		
12	18	LED Lamp	Home	1	54.69	10	0	Reese Okafor	Champion	Social	LED Lamp	Home		
13	26	Shampoo Bar	Beauty	3	29.61	10	0	Reese Costa	Champion	Direct	Shampoo Bar	Beauty		
14	31	Resistance Bands	Sports	2	37.72	10	0	Mateo Patel	Loyal	Social	Resistance Bands	Sports		
15	12	Bluetooth Speaker	Electronics	1	110.84	10	0	Noel Novak	Loyal	Organic	Bluetooth Speaker	Electronics		
16	19	Cutting Board	Home	1	11.13	20	0	Morgan Nguyen	New	Social	Cutting Board	Home		
17	13	Webcam	Electronics	1	60.48	0	0	Jamie Costa	Champion	Paid Search	Webcam	Electronics		

	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	product_id	product	category	quantity	unit_price	Revenue	discount_pct	Revenue after	returned	cleaned Customer	Cleaned	Cleaned Channel	cleaned Product	Cleaned Category
2	28	SPF 50 Lotion	Beauty	1	23.27	23.27	5	22.1065	0	Hugo Brooks	Loyal	Paid Search	SPF 50 Lotion	Beauty
3	5	Socks 3-Pack	Apparel	2	58.34	116.68	0	116.68	0	Owen Nguyen	Regular	Organic	Socks 3-Pack	Apparel
4	31	Resistance Bands	Sports	1	37.72	37.72	25	28.29	0	Lila Mendez	Regular	Direct	Resistance Bands	Sports
5	5	Socks 3-Pack	Apparel	2	58.34	116.68	10	105.012	0	Jamie Romano	Loyal	Organic	Socks 3-Pack	Apparel
6	32	Foam Roller	Sports	1	51.36	51.36	10	46.224	0	Tessa Okafor	Loyal	Referral	Foam Roller	Sports
7	29	Yoga Mat	Sports	1	31.47	31.47	0	31.47	0	Cole Okafor	Champion	Referral	Yoga Mat	Sports
8	1	Crew Tee	Apparel	3	67.83	203.49	10	183.141	0	Owen Petrov	Champion	Paid Search	Crew Tee	Apparel
9	22	Scented Candle	Home	1	33.84	33.84	0	33.84	0	Casey Costa	New	Organic	Scented Candle	Home
10	40	Art Kit	Toys	1	21.25	21.25	0	21.25	0	Priya Sullivan	Regular	Email	Art Kit	Toys
11	34	Running Belt	Sports	1	50.13	50.13	0	50.13	0	Devon Sullivan	Loyal	Referral	Running Belt	Sports
12	18	LED Lamp	Home	1	54.69	54.69	10	49.221	0	Reese Okafor	Champion	Social	LED Lamp	Home
13	26	Shampoo Bar	Beauty	3	29.61	88.83	10	79.947	0	Reese Costa	Champion	Direct	Shampoo Bar	Beauty
14	31	Resistance Bands	Sports	2	37.72	75.44	10	67.896	0	Mateo Patel	Loyal	Social	Resistance Bands	Sports
15	12	Bluetooth Speaker	Electronics	1	110.84	110.84	10	99.756	0	Noel Novak	Loyal	Organic	Bluetooth Speaker	Electronics
16	19	Cutting Board	Home	1	11.13	11.13	20	8.904	0	Morgan Nguyen	New	Social	Cutting Board	Home
17	13	Webcam	Electronics	1	60.48	60.48	0	60.48	0	Jamie Costa	Champion	Paid Search	Webcam	Electronics
18	12	Bluetooth Speaker	Electronics	3	110.84	332.52	0	332.52	0	Hugo Flynn	Loyal	Email	Bluetooth Speaker	Electronics
19	36	Puzzle 500pc	Toys	1	44.06	44.06	0	44.06	0	Reese Carter	Regular	Paid Search	Puzzle 500pc	Toys
20	13	Webcam	Electronics	1	60.48	60.48	0	60.48	0	Drew Kowalski	New	Email	Webcam	Electronics
21	2	Hoodie	Apparel	1	24.05	24.05	0	24.05	0	Tessa Osei	Loyal	Organic	Hoodie	Apparel
22	23	Vitamin C Serum	Beauty	1	20.25	20.25	0	20.25	0	Iris Bauer	Regular	Social	Vitamin C Serum	Beauty
23	25	Sheet Mask	Beauty	1	5.03	5.03	25	3.7725	1	Maya Fischer	Loyal	Email	Sheet Mask	Beauty
24	10	USB-C Cable	Electronics	1	29.62	29.62	10	26.658	0	Nora Yamada	Loyal	Organic	USB-C Cable	Electronics
25	34	Running Belt	Sports	1	50.13	50.13	0	50.13	0	Maya Park	Regular	Paid Search	Running Belt	Sports

	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	customer_id	customer	segment	channel	product_id	product	category	quantity	unit_price	Revenue	discount_pct	Column1	turned	cleaned Customer	Cleaned
2	20	Hugo Brooks	Loyal	Paid Search	28	SPF 50 Lotion	Beauty	1	23.27	23.27	5		0	Hugo Brooks	Loyal
3	41	Owen Nguyen	Regular	Organic	5	Socks 3-Pack	Apparel	2	58.34	116.68	0		0	Owen Nguyen	Regular
4	86	Lila Mendez	Regular	Direct	31	Resistance Bands	Sports	1	37.72	37.72	25		0	Lila Mendez	Regular
5	97	Jamie Romano	Loyal	Organic	5	Socks 3-Pack	Apparel	2	58.34	116.68	10		0	Jamie Romano	Loyal
6	108	Tessa Okafor	Loyal	Referral	32	Foam Roller	Sports	1	51.36	51.36	10		0	Tessa Okafor	Loyal
7	113	Cole Okafor	Champion	Referral	29	Yoga Mat	Sports	1	31.47	31.47	0		0	Cole Okafor	Champion
8	115	Owen Petrov	Champion	Paid Search	1	Crew Tee	Apparel	3	67.83	203.49	10		0	Owen Petrov	Champion
9	119	Casey Costa	New	Organic	22	Scented Candle	Home	1	33.84	33.84	0		0	Casey Costa	New
10	138	Priya Sullivan	Regular	Email	40	Art Kit	Toys	1	21.25	21.25	0		0	Priya Sullivan	Regular
11	141	Devon Sullivan	Loyal	Referral	34	Running Belt	Sports	1	50.13	50.13	0		0	Devon Sullivan	Loyal
12	165	Reese Okafor	Champion	Social	18	LED Lamp	Home	1	54.69	54.69	10		0	Reese Okafor	Champion
13	343	Reese Costa	Champion	Direct	26	Shampoo Bar	Beauty	3	29.61	88.83	10		0	Reese Costa	Champion
14	379	Mateo Patel	Loyal	Social	31	Resistance Bands	Sports	2	37.72	75.44	10		0	Mateo Patel	Loyal
15	395	Noel Novak	Loyal	Organic	12	Bluetooth Speaker	Electronics	1	110.84	110.84	10		0	Noel Novak	Loyal
16	447	Morgan Nguyen	New	Social	19	Cutting Board	Home	1	11.13	11.13	20		0	Morgan Nguyen	New
17	470	Jamie Costa	Champion	Paid Search	13	Webcam	Electronics	1	60.48	60.48	0		0	Jamie Costa	Champion
18	568	Hugo Flynn	Loyal	Email	12	Bluetooth Speaker	Electronics	3	110.84	332.52	0		0	Hugo Flynn	Loyal
19	593	Reese Carter	Regular	Paid Search	36	Puzzle 500pc	Toys	1	44.06	44.06	0		0	Reese Carter	Regular
20	612	Drew Kowalski	New	Email	13	Webcam	Electronics	1	60.48	60.48	0		0	Drew Kowalski	New
21	644	Tessa Osei	Loyal	Organic	2	Hoodie	Apparel	1	24.05	24.05	0		0	Tessa Osei	Loyal
22	692	Iris Bauer	Regular	Social	23	Vitamin C Serum	Beauty	1	20.25	20.25	0		0	Iris Bauer	Regular
23	772	Maya Fischer	Loyal	Email	25	Sheet Mask	Beauty	1	5.03	5.03	25		1	Maya Fischer	Loyal
24	801	Nora Yamada	Loyal	Organic	10	USB-C Cable	Electronics	1	29.62	29.62	10		0	Nora Yamada	Loyal

	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	product_id	product	category	quantity	unit_price	Revenue	discount_pct	Revenue after	returned	cleaned Customer	Cleaned	Cleaned Channel	cleaned Product	Cleaned Category	
2	28	SPF 50 Lotion	Beauty	1	23.27	23.27	5	22.1065	0	Hugo Brooks	Loyal	Paid Search	SPF 50 Lotion	Beauty	
3	5	Socks 3-Pack	Apparel	2	58.34	116.68	0	116.68	0	Owen Nguyen	Regular	Organic	Socks 3-Pack	Apparel	
4	31	Resistance Bands	Sports	1	37.72	37.72	25	28.29	0	Lila Mendez	Regular	Direct	Resistance Bands	Sports	
5	5	Socks 3-Pack	Apparel	2	58.34	116.68	10	105.012	0	Jamie Romano	Loyal	Organic	Socks 3-Pack	Apparel	
6	32	Foam Roller	Sports	1	51.36	51.36	10	46.224	0	Tessa Okafor	Loyal	Referral	Foam Roller	Sports	
7	29	Yoga Mat	Sports	1	31.47	31.47	0	31.47	0	Cole Okafor	Champion	Referral	Yoga Mat	Sports	
8	1	Crew Tee	Apparel	3	67.83	203.49	10	183.141	0	Owen Petrov	Champion	Paid Search	Crew Tee	Apparel	
9	22	Scented Candle	Home	1	33.84	33.84	0	33.84	0	Casey Costa	New	Organic	Scented Candle	Home	
10	40	Art Kit	Toys	1	21.25	21.25	0	21.25	0	Priya Sullivan	Regular	Email	Art Kit	Toys	
11	34	Running Belt	Sports	1	50.13	50.13	0	50.13	0	Devon Sullivan	Loyal	Referral	Running Belt	Sports	
12	18	LED Lamp	Home	1	54.69	54.69	10	49.221	0	Reese Okafor	Champion	Social	LED Lamp	Home	
13	26	Shampoo Bar	Beauty	3	29.61	88.83	10	79.947	0	Reese Costa	Champion	Direct	Shampoo Bar	Beauty	
14	31	Resistance Bands	Sports	2	37.72	75.44	10	67.896	0	Mateo Patel	Loyal	Social	Resistance Bands	Sports	
15	12	Bluetooth Speaker	Electronics	1	110.84	110.84	10	99.756	0	Noel Novak	Loyal	Organic	Bluetooth Speaker	Electronics	
16	19	Cutting Board	Home	1	11.13	11.13	20	8.904	0	Morgan Nguyen	New	Social	Cutting Board	Home	
17	13	Webcam	Electronics	1	60.48	60.48	0	60.48	0	Jamie Costa	Champion	Paid Search	Webcam	Electronics	
18	12	Bluetooth Speaker	Electronics	3	110.84	332.52	0	332.52	0	Hugo Flynn	Loyal	Email	Bluetooth Speaker	Electronics	
19	36	Puzzle 500pc	Toys	1	44.06	44.06	0	44.06	0	Reese Carter	Regular	Paid Search	Puzzle 500pc	Toys	
20	13	Webcam	Electronics	1	60.48	60.48	0	60.48	0	Drew Kowalski	New	Email	Webcam	Electronics	
21	2	Hoodie	Apparel	1	24.05	24.05	0	24.05	0	Tessa Osei	Loyal	Organic	Hoodie	Apparel	
22	23	Vitamin C Serum	Beauty	1	20.25	20.25	0	20.25	0	Iris Bauer	Regular	Social	Vitamin C Serum	Beauty	
23	25	Sheet Mask	Beauty	1	5.03	5.03	25	3.7725	1	Maya Fischer	Loyal	Email	Sheet Mask	Beauty	
24	10	USB-C Cable	Electronics	1	29.62	29.62	10	26.658	0	Nora Yamada	Loyal	Organic	USB-C Cable	Electronics	
25	34	Running Belt	Sports	1	50.13	50.13	0	50.13	0	Maya Park	Regular	Paid Search	Running Belt	Sports	
26	15	Smart Plug	Electronics	3	8.53	25.59	0	25.59	0	Sasha Diaz	Regular	Paid Search	Smart Plug	Electronics	

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	order_date	order_id	customer_id	customer	segment	channel	product_id	product	category	quantity	unit_price	discount_pct	returned			
2	2025-08-02	1001	20	Hugo Brooks	Loyal	Paid Search	28	SPF 50 Loti	Beauty	1	23.27		5	0		
3	2025-08-02	1002	41	Owen Nguye	Regular	Organic	5	Socks 3-Pac	Apparel	2	58.34	0	0			
4	2025-08-02	1002	41	Owen Nguye	Regular	Organic	32	Foam Rolle	Sports	1	51.36	0	0			
5	2025-08-02	1002	41	Owen Nguye	Regular	Organic	15	Smart Plug	Electronics	1	8.53		0	0		
6	2025-08-02	1003	86	Lila Mendez	Regular	Direct	31	Resistance	Sports	1	37.72	25	0			
7	2025-08-02	1004	97	Jamie Romar	Loyal	Organic	5	Socks 3-Pac	Apparel	2	58.34		10	0		
8	2025-08-02	1005	108	Tessa Okafo	Loyal	Referral	32	Foam Rolle	Sports	1	51.36	10	0			
9	2025-08-02	1005	108	Tessa Okafo	Loyal	Referral	22	Scented Ca	Home	3	33.84		5	0		
10	2025-08-02	1005	108	Tessa Okafo	Loyal	Referral	8	Chino Pants	Apparel	2	25.32		5	0		
11	2025-08-02	1005	108	Tessa Okafo	Loyal	Referral	29	Yoga Mat	Sports	1	31.47		5	0		
12	2025-08-02	1006	113	Cole Okafo	Champion	Referral	29	Yoga Mat	Sports	1	31.47		0	0		
13	2025-08-02	1006	113	Cole Okafo	Champion	Referral	33	Jump Rope	Sports	1	64.12		0	0		
14	2025-08-02	1006	113	Cole Okafo	Champion	Referral	35	Building Bl	Toys	2	20.6		0	0		
15	2025-08-02	1006	113	Cole Okafo	Champion	Referral	18	LED Lamp	Home	1	54.69	10	1			
16	2025-08-02	1007	115	Owen Petrov	Champion	Paid Search	1	Crew Tee	Apparel	3	67.83	10	0			
17	2025-08-02	1008	119	Casey Costa	New	Organic	22	Scented Ca	Home	1	33.84		0	0		
18	2025-08-02	1009	138	Priya Sullivan	Regular	Email	40	Art Kit	Toys	1	21.25		0	0		
19	2025-08-02	1009	138	Priya Sullivan	Regular	Email	3	Denim Jack	Apparel	1	17.82		0	0		
20	2025-08-02	1009	138	Priya Sullivan	Regular	Email	18	LED Lamp	Home	2	54.69		0	0		
21	2025-08-02	1009	138	Priya Sullivan	Regular	Email	30	Water Bott	Sports	1	45.18		0	0		
22	2025-08-02	1010	141	Devon Sulliv	Loyal	Referral	34	Running Be	Sports	1	50.13		0	0		
23	2025-08-02	1011	165	Reese Okafo	Champion	Social	18	LED Lamp	Home	1	54.69	10	0			
24	2025-08-02	1011	165	Reese Okafo	Champion	Social	33	Jump Rope	Sports	3	64.12		15	0		
25	2025-08-02	1012	343	Reese Costa	Champion	Direct	26	Shampoo B	Beauty	3	29.61		10	0		
26	2025-08-02	1013	379	Mateo Patel	Loyal	Social	31	Resistance	Sports	2	37.72		10	0		
27	2025-08-02	1014	395	Noel Novak	Loyal	Organic	12	Bluetooth S	Electronics	1	110.84	10	0			
28	2025-08-02	1014	395	Noel Novak	Loyal	Organic	7	Rain Shell	Apparel	1	35.17	15	0			
29	2025-08-02	1015	447	Morgan Ngu	New	Social	19	Cutting Boa	Home	1	11.13	20	0			
30	2025-08-02	1015	447	Morgan Ngu	New	Social	19	Cutting Boa	Home	1	11.13	15	0			
31	2025-08-02	1016	470	Jamie Costa	Champion	Paid Search	13	Webcam	Electronics	1	60.48		0	0		
32	2025-08-02	1017	568	Hugo Flynn	Loyal	Email	12	Bluetooth S	Electronics	3	110.84		0	0		
33	2025-08-02	1017	568	Hugo Flynn	Loyal	Email	26	Shampoo B	Beauty	1	29.61	10	0			
34	2025-08-02	1018	593	Reese Carter	Regular	Paid Search	36	Puzzle 500p	Toys	1	44.06		0	0		
35	2025-08-02	1018	593	Reese Carter	Regular	Paid Search	7	Rain Shell	Apparel	2	35.17		0	1		
36	2025-08-02	1018	593	Reese Carter	Regular	Paid Search	13	Webcam	Electronics	1	60.48		0	0		

product_id	Product Name	Category Name	Unit Price
1	Crew Tee	Apparel	67.83
2	Hoodie	Apparel	24.05
3	Denim Jacket	Apparel	17.82
4	Joggers	Apparel	76.5
5	Socks 3-Pack	Apparel	58.34
6	Beanie	Apparel	54.22
7	Rain Shell	Apparel	35.17
8	Chino Pants	Apparel	25.32
9	Wireless Earbuds	Electronics	78.18
10	USB-C Cable	Electronics	29.62
11	Phone Case	Electronics	8.61
12	Bluetooth Speaker	Electronics	110.84
13	Webcam	Electronics	60.48
14	Power Bank	Electronics	46.23
15	Smart Plug	Electronics	8.53
16	Throw Pillow	Home	10.61
17	Ceramic Mug	Home	38.42
18	LED Lamp	Home	54.69
19	Cutting Board	Home	11.13
20	Towel Set	Home	55.04
21	Storage Bin	Home	25.45
22	Scented Candle	Home	33.84
23	Vitamin C Serum	Beauty	20.25
24	Lip Balm	Beauty	20.91
25	Sheet Mask	Beauty	5.03
26	Shampoo Bar	Beauty	29.61
27	Hand Cream	Beauty	15.73
28	SPF 50 Lotion	Beauty	23.27
29	Yoga Mat	Sports	31.47
30	Water Bottle	Sports	45.18
31	Resistance Bands	Sports	37.72
32	Foam Roller	Sports	51.36
33	Jump Rope	Sports	64.12
34	Running Belt	Sports	50.13

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1															
2	Product	Sum of quantity	Customer	Sum of quantity	Product	Count of order_id	Customer	Count of order_id							
3	Art Kit	184	Avery Bauer	9	Art Kit	130	Avery Bauer	8							
4	Beanie	189	Avery Costa	12	Beanie	149	Avery Costa	10							
5	Bluetooth Speaker	215	Avery Diaz	10	Bluetooth Speaker	151	Avery Diaz	5							
6	Board Game	176	Avery Flynn	9	Board Game	128	Avery Flynn	6							
7	Building Blocks	205	Avery Hassan	3	Building Blocks	143	Avery Hassan	3							
8	Ceramic Mug	223	Avery Khan	4	Ceramic Mug	152	Avery Khan	3							
9	Chino Pants	194	Avery Kowalski	15	Chino Pants	142	Avery Kowalski	12							
10	Crew Tee	195	Avery Larsen	21	Crew Tee	142	Avery Larsen	13							
11	Cutting Board	187	Avery Nguyen	4	Cutting Board	138	Avery Nguyen	4							
12	Denim Jacket	186	Avery Novak	9	Denim Jacket	130	Avery Novak	4							
13	Foam Roller	194	Avery Okafor	13	Foam Roller	133	Avery Okafor	11							
14	Hand Cream	186	Avery Patel	5	Hand Cream	138	Avery Patel	4							
15	Hoodie	187	Avery Petrov	7	Hoodie	139	Avery Petrov	3							
16	Joggers	204	Avery Reyes	7	Joggers	152	Avery Reyes	5							
17	Jump Rope	171	Avery Sullivan	4	Jump Rope	125	Avery Sullivan	2							
18	LED Lamp	168	Casey Bauer	6	LED Lamp	119	Casey Bauer	3							
19	Lip Balm	173	Casey Brooks	3	Lip Balm	129	Casey Brooks	3							
20	Phone Case	193	Casey Carter	21	Phone Case	144	Casey Carter	17							

Highlighted Returned items in red.

Highlighted Gross Pay = 0 for returned items in red.

Highlighted Profit = 0 in red.

Highlighted Profit > 0 in green.

Used Conditional Formatting for quick analysis and better visualization.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	order_date	order_id	Customer_id	product_id	quantity	Unit_price	Gross Sal	discount_pct	Revenue	returned	Net Sales	Cleaned Customer	Cleaned Segment	Cleaned Channel	Channel
2	02-08-2025 00:00	1001	20	28	1	23.27	23.27		22.1065	0	22.1065	Hugo Brooks	Loyal	Paid Search	SP
3	02-08-2025 00:00	1002	41	5	2	58.34	116.68		116.68	0	116.68	Owen Nguyen	Regular	Organic	So
4	02-08-2025 00:00	1003	86	31	1	37.72	37.72		28.29	0	28.29	Lilla Mendez	Regular	Direct	Re
5	02-08-2025 00:00	1004	97	5	2	58.34	116.68		105.012	0	105.012	Jamie Romano	Loyal	Organic	So
6	02-08-2025 00:00	1005	108	32	1	51.36	51.36		46.224	0	46.224	Tessa Okafor	Loyal	Referral	Re
7	02-08-2025 00:00	1006	113	29	1	31.47	31.47		31.47	0	31.47	Cole Okafor	Champion	Referral	Yo
8	02-08-2025 00:00	1007	115	1	3	67.83	203.49		183.141	0	183.141	Owen Petrov	Champion	Paid Search	Ci
9	02-08-2025 00:00	1008	119	22	1	33.84	33.84		33.84	0	33.84	Casey Costa	New	Organic	So
10	02-08-2025 00:00	1009	138	40	1	21.25	21.25		21.25	0	21.25	Priya Sullivan	Regular	Email	Ar
11	02-08-2025 00:00	1010	141	34	1	50.13	50.13		50.13	0	50.13	Devon Sullivan	Loyal	Referral	Ru
12	02-08-2025 00:00	1011	165	18	1	54.69	54.69		49.221	0	49.221	Reese Okafor	Champion	Social	LE
13	02-08-2025 00:00	1012	343	26	3	29.61	88.83		79.947	0	79.947	Reese Costa	Champion	Direct	Sh
14	02-08-2025 00:00	1013	379	31	2	37.72	75.44		67.896	0	67.896	Mateo Patel	Loyal	Social	Re
15	02-08-2025 00:00	1014	395	12	1	110.84	110.84		99.756	0	99.756	Noel Novak	Loyal	Organic	Bl
16	02-08-2025 00:00	1015	447	19	1	11.13	11.13		8.904	0	8.904	Morgan Nguyen	New	Social	Cl
17	02-08-2025 00:00	1016	470	13	1	60.48	60.48		60.48	0	60.48	Jamie Costa	Champion	Paid Search	W
18	02-08-2025 00:00	1017	568	12	3	110.84	332.52		332.52	0	332.52	Hugo Flynn	Loyal	Email	Bl
19	02-08-2025 00:00	1018	593	36	1	44.06	44.06		44.06	0	44.06	Reese Carter	Regular	Paid Search	Pu
20	02-08-2025 00:00	1019	612	13	1	60.48	60.48		60.48	0	60.48	Drew Kowalski	New	Email	W
21	02-08-2025 00:00	1020	644	2	1	24.05	24.05		24.05	0	24.05	Tessa Osei	Loyal	Organic	Hg
22	02-08-2025 00:00	1021	692	23	1	20.25	20.25		20.25	0	20.25	Iris Bauer	Regular	Regular	W
23	02-08-2025 00:00	1022	772	25	1	5.03	5.03		3.7725	1	0	Maya Fischer	Loyal	Email	Sh
24	02-08-2025 00:00	1023	801	10	1	29.62	29.62		26.658	0	26.658	Nora Yamada	Loyal	Organic	US
25	02-08-2025 00:00	1024	823	34	1	50.13	50.13		50.13	0	50.13	Maya Park	Regular	Paid Search	Ru
26	02-08-2025 00:00	1025	852	15	3	8.53	25.59		25.59	0	25.59	Sasha Diaz	Regular	Paid Search	Sn
27	02-08-2025 00:00	1026	887	30	1	45.18	45.18		42.921	0	42.921	Mateo Novak	Champion	Referral	W
28	02-08-2025 00:00	1027	905	13	1	60.48	60.48		60.48	0	60.48	Tessa Reid	Loyal	Email	W
29	02-08-2025 00:00	1028	906	30	1	45.18	45.18		42.921	0	42.921	Noel Flynn	Champion	Social	W
30	02-08-2025 00:00	1029	942	10	1	29.62	29.62		29.62	0	29.62	Maya Kowalski	Loyal	Organic	US
31	02-08-2025 00:00	1030	970	38	1	15.37	15.37		15.37	0	15.37	Ezra Sullivan	Regular	Organic	Bo

Summary

- Cleaned the e-commerce order data using Excel and Power Query.
- Removed duplicates, handled missing values, and standardized data formats.
- Created separate Customer and Product tables.
- Used XLOOKUP to retrieve customer table and product table
- Created Pivot Tables to summarize order and quantity data for customer and product
- Prepared the dataset for dashboard creation in Power BI.

Observations

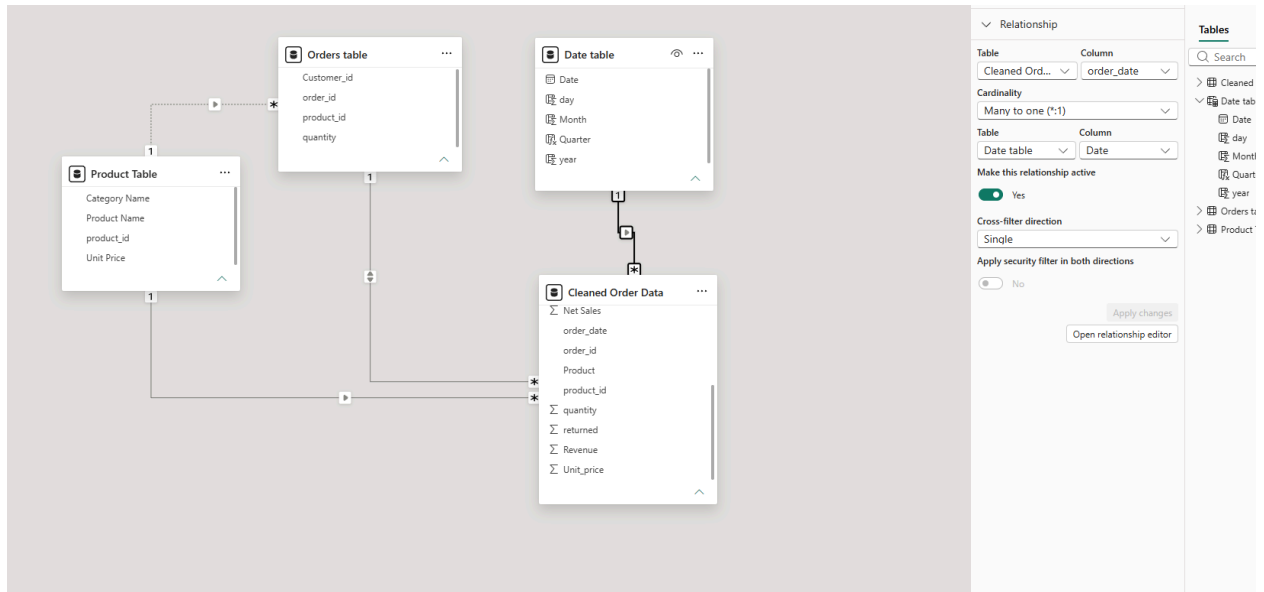
- The data became clean, accurate, and easy to analyze after preprocessing.
- Separating Customer and Product tables reduced duplicate data.
- XLOOKUP made it easy to retrieve related information.
- Pivot Tables quickly summarized customer and product performance.
- Pivot Tables to summarize the data and analyze customer orders, product performance, and total quantities. They helped me quickly identify top customers, top-selling products, and overall order trends without writing formulas.
- The structured dataset is ready for creating interactive dashboards and business reports.
- Initial analysis showed that the dataset contained order transactions with repeated customer and product information.

7. Data Modelling and DAX (Power BI)

Power BI data model was designed using a Star Schema, Cleaned Order Data table as the central fact table and the Product Table and Date Table as dimension tables.

One-to-many relationships with single-direction filtering applied here. efficient data analysis, accurate aggregations, and improved report performance.

And created date table for to find YTD sale



Report view					Date table = CALENDAR(MIN('Cleaned Order Data'[order_date]),MAX('Cleaned Order Data'[order_date]))				
Date	Month	Quarter	day	year					
02-08-2025 00:00:00	8	Q3	2	2025					
03-08-2025 00:00:00	8	Q3	3	2025					
04-08-2025 00:00:00	8	Q3	4	2025					
05-08-2025 00:00:00	8	Q3	5	2025					
06-08-2025 00:00:00	8	Q3	6	2025					
07-08-2025 00:00:00	8	Q3	7	2025					
08-08-2025 00:00:00	8	Q3	8	2025					
09-08-2025 00:00:00	8	Q3	9	2025					
10-08-2025 00:00:00	8	Q3	10	2025					
11-08-2025 00:00:00	8	Q3	11	2025					
12-08-2025 00:00:00	8	Q3	12	2025					
13-08-2025 00:00:00	8	Q3	13	2025					
14-08-2025 00:00:00	8	Q3	14	2025					
15-08-2025 00:00:00	8	Q3	15	2025					
16-08-2025 00:00:00	8	Q3	16	2025					
17-08-2025 00:00:00	8	Q3	17	2025					
18-08-2025 00:00:00	8	Q3	18	2025					
19-08-2025 00:00:00	8	Q3	19	2025					
20-08-2025 00:00:00	8	Q3	20	2025					
21-08-2025 00:00:00	8	Q3	21	2025					
22-08-2025 00:00:00	8	Q3	22	2025					
23-08-2025 00:00:00	8	Q3	23	2025					
24-08-2025 00:00:00	8	Q3	24	2025					
25-08-2025 00:00:00	8	Q3	25	2025					
26-08-2025 00:00:00	8	Q3	26	2025					
27-08-2025 00:00:00	8	Q3	27	2025					
28-08-2025 00:00:00	8	Q3	28	2025					
29-08-2025 00:00:00	8	Q3	29	2025					
30-08-2025 00:00:00	8	Q3	30	2025					

Created a target as 120% of current sales

Using Net sales Target = Total net sales * 1.20

Sheet1		Formatting		Properties		Measure		Calculations								Data	
1 Net Sales Target = [Total Net Sales]*1.20																	
order_date	order_id	customer_id	product_id	quantity	Unit_price	Gross Sale	discount_pct	Revenue	returned	Net Sales	Customer	Segment	Channel	Product	Category	Profit	
02-08-2025 00:00:00	1001	20	28	1	23.27	23.27	5	22.1065	0	22.1065	Hugo Brooks	Loyal	Paid Search	SPF 50 Lotion	Beauty	1.1635	Search
02-08-2025 00:00:00	1002	41	5	2	58.34	116.68	0	116.68	0	116.68	Owen Nguyen	Regular	Organic	Socks 3-Pack	Apparel	0	Cleaned Order Id
02-08-2025 00:00:00	1003	86	31	1	37.72	37.72	25	28.29	0	28.29	Lila Mendez	Regular	Direct	Resistance Bands	Sports	9.43	Customer
02-08-2025 00:00:00	1004	97	5	2	58.34	116.68	10	105.012	0	105.012	Jamie Romano	Loyal	Organic	Socks 3-Pack	Apparel	11.668	Segment
02-08-2025 00:00:00	1005	108	32	1	51.36	51.36	10	46.224	0	46.224	Tessa Okafor	Loyal	Referral	Foam Roller	Sports	5.136	Avg Sale
02-08-2025 00:00:00	1006	113	29	1	31.47	31.47	0	31.47	0	31.47	Cole Okafor	Champion	Referral	Yoga Mat	Sports	0	Category
02-08-2025 00:00:00	1007	115	1	3	67.83	203.49	10	183.141	0	183.141	Owen Petrov	Champion	Paid Search	Crew Tee	Apparel	20.349	Channel
02-08-2025 00:00:00	1008	119	22	1	33.84	33.84	0	33.84	0	33.84	Casey Costa	New	Organic	Scented Candle	Home	0	Customer_id
02-08-2025 00:00:00	1009	138	40	1	21.25	21.25	0	21.25	0	21.25	Priya Sullivan	Regular	Email	Art Kit	Toys	0	discount_pct
02-08-2025 00:00:00	1010	141	34	1	50.13	50.13	0	50.13	0	50.13	Devon Sullivan	Loyal	Referral	Running Belt	Sports	0	Gross Sale
02-08-2025 00:00:00	1011	165	18	1	54.69	54.69	10	49.221	0	49.221	Reese Okafor	Champion	Social	LED Lamp	Home	5.469	Max Sale
02-08-2025 00:00:00	1012	343	26	3	29.61	88.83	10	79.947	0	79.947	Reese Costa	Champion	Direct	Shampoo Bar	Beauty	8.883	Min Sale
02-08-2025 00:00:00	1013	379	31	2	37.72	75.44	10	67.896	0	67.896	Mateo Patel	Loyal	Social	Resistance Bands	Sports	7.544	Net Sales
02-08-2025 00:00:00	1014	395	12	1	110.84	110.84	10	99.756	0	99.756	Noel Novak	Loyal	Organic	Bluetooth Speaker	Electronics	11.084	Net Sales Target
02-08-2025 00:00:00	1015	447	19	1	11.13	11.13	20	8.904	0	8.904	Morgan Nguyen	New	Social	Cutting Board	Home	2.226	Net Sales Target
02-08-2025 00:00:00	1016	470	13	1	60.48	60.48	0	60.48	0	60.48	Jamie Costa	Champion	Paid Search	Webcam	Electronics	0	order_date
02-08-2025 00:00:00	1017	568	12	3	110.84	332.52	0	332.52	0	332.52	Hugo Flynn	Loyal	Email	Bluetooth Speaker	Electronics	0	order_id
02-08-2025 00:00:00	1018	593	36	1	44.06	44.06	0	44.06	0	44.06	Reese Carter	Regular	Paid Search	Puzzle 500pc	Toys	0	
02-08-2025 00:00:00	1019	612	13	1	60.48	60.48	0	60.48	0	60.48	Drew Kowalski	New	Email	Webcam	Electronics	0	

Create a Target as 130% of Current Profit

Using Profit Target = Total Profit * 1.30

Profit Target = [Total Profit]*1.30																
Properties																
Calculations																
Data																
order_date	order_id	customer_id	product_id	quantity	unit_price	Gross Sale	discount_pct	Revenue	returned	Net Sales	Customer	Segment	Channel	Product	Category	Profit
2025-00-00-00	1001	20	28	1	23.27	23.27	5	22.1065	0	22.1065	Hugo Brooks	Loyal	Paid Search	SPF 50 Lotion	Beauty	1.1635
2025-00-00-00	1002	41	5	2	58.34	116.68	0	116.68	0	116.68	Owen Nguyen	Regular	Organic	Socks 3-Pack	Apparel	0
2025-00-00-00	1003	86	31	1	37.72	37.72	25	28.29	0	28.29	Lila Mendez	Regular	Direct	Resistance Bands	Sports	9.43
2025-00-00-00	1004	97	5	2	58.34	116.68	10	105.012	0	105.012	Jamie Romano	Loyal	Organic	Socks 3-Pack	Apparel	11.668
2025-00-00-00	1005	108	32	1	51.36	51.36	10	46.224	0	46.224	Tessa Okafor	Loyal	Referral	Foam Roller	Sports	5.136
2025-00-00-00	1006	113	29	1	31.47	31.47	0	31.47	0	31.47	Cole Okafor	Champion	Referral	Yoga Mat	Sports	0
2025-00-00-00	1007	115	1	3	67.83	203.49	10	183.141	0	183.141	Owen Petrov	Champion	Paid Search	Crew Tee	Apparel	20.349
2025-00-00-00	1008	119	22	1	33.84	33.84	0	33.84	0	33.84	Casey Costa	New	Organic	Scented Candle	Home	0
2025-00-00-00	1009	138	40	1	21.25	21.25	0	21.25	0	21.25	Priya Sullivan	Regular	Email	Art Kit	Toys	0
2025-00-00-00	1010	141	34	1	50.13	50.13	0	50.13	0	50.13	Devon Sullivan	Loyal	Referral	Running Belt	Sports	0
2025-00-00-00	1011	165	18	1	54.69	54.69	10	49.221	0	49.221	Reese Okafor	Champion	Social	LED Lamp	Home	5.469
2025-00-00-00	1012	343	26	3	29.61	88.83	10	79.947	0	79.947	Reese Costa	Champion	Direct	Shampoo Bar	Beauty	8.883
2025-00-00-00	1013	379	31	2	37.72	75.44	10	67.896	0	67.896	Mateo Patel	Loyal	Social	Resistance Bands	Sports	7.544
2025-00-00-00	1014	395	12	1	110.84	110.84	10	99.756	0	99.756	Noel Novak	Loyal	Organic	Bluetooth Speaker	Electronics	11.084
2025-00-00-00	1015	447	19	1	11.13	11.13	20	8.904	0	8.904	Morgan Nguyen	New	Social	Cutting Board	Home	2.226
2025-00-00-00	1016	470	13	1	60.48	60.48	0	60.48	0	60.48	Jamie Costa	Champion	Paid Search	Webcam	Electronics	0
2025-00-00-00	1017	568	12	3	110.84	332.52	0	332.52	0	332.52	Hugo Flynn	Loyal	Email	Bluetooth Speaker	Electronics	0
2025-00-00-00	1018	593	36	1	44.06	44.06	0	44.06	0	44.06	Reese Carter	Regular	Paid Search	Puzzle 500pc	Toys	0
2025-00-00-00	1019	612	13	1	60.48	60.48	0	60.48	0	60.48	Drew Kowalski	New	Email	Webcam	Electronics	0
2025-00-00-00	1020	644	2	1	24.05	24.05	0	24.05	0	24.05	Tessa Osei	Loyal	Organic	Hoodie	Apparel	0
2025-00-00-00	1021	692	23	1	20.25	20.25	0	20.25	0	20.25	Iris Bauer	Regular	Social	Vitamin C Serum	Beauty	0
2025-00-00-00	1022	772	25	1	5.03	5.03	25	3.7725	1	0	Maya Fischer	Loyal	Email	Sheet Mask	Beauty	5.03
2025-00-00-00	1023	801	10	1	29.62	29.62	10	26.658	0	26.658	Nora Yamada	Loyal	Organic	USB-C Cable	Electronics	2.962
2025-00-00-00	1024	823	34	1	50.13	50.13	0	50.13	0	50.13	Maya Park	Regular	Paid Search	Smart Plug	Sports	0
2025-00-00-00	1025	852	15	3	8.53	25.59	0	25.59	0	25.59	Sasha Diaz	Regular	Paid Search	Smart Plug	Electronics	0

Created a Target as 125% of Current Quantity

Using Quantity Target = total Quantity * 1.25

Name		Formatting		Properties		measure measure		Calculations								Data		
1		Quantity Target = [Total Quantity]*1.25																
date	order_id	customer_id	product_id	quantity	unit_price	Gross Sale	discount_pct	Revenue	returned	Net Sales	Customer	Segment	Channel	Product	Category	Profit		
15 00:00:00	1001	20	28	1	23.27	23.27		5	22.1065	0	22.1065	Hugo Brooks	Loyal	Paid Search	SPF 50 Lotion	Beauty	1.1635	Q Search
15 00:00:00	1002	41	5	2	58.34	116.68		0	116.68	0	116.68	Owen Nguyen	Regular	Organic	Socks 3-Pack	Apparel	0	Cleaned Order Data
15 00:00:00	1003	86	31	1	37.72	37.72	25	28.29	0	28.29	Lila Mendez	Regular	Direct	Resistance Bands	Sports	9.43	Customer	
15 00:00:00	1004	97	5	2	58.34	116.68		10	105.012	0	105.012	Jamie Romano	Loyal	Organic	Socks 3-Pack	Apparel	11.668	Segment
15 00:00:00	1005	108	32	1	51.36	51.36		10	46.224	0	46.224	Tessa Okafor	Loyal	Referral	Foam Roller	Sports	5.136	Av Sale
15 00:00:00	1006	113	29	1	31.47	31.47		0	31.47	0	31.47	Cole Okafor	Champion	Referral	Yoga Mat	Sports	0	Category
15 00:00:00	1007	115	1	3	67.83	203.49	10	183.141	0	183.141	Owen Petrov	Champion	Paid Search	Crew Tee	Apparel	20.349	Channel	
15 00:00:00	1008	119	22	1	33.84	33.84		0	33.84	0	33.84	Casey Costa	New	Organic	Scented Candle	Home	0	Customer_id
15 00:00:00	1009	138	40	1	21.25	21.25		0	21.25	0	21.25	Priya Sullivan	Regular	Email	Art Kit	Toys	0	discount_pct
15 00:00:00	1010	141	34	1	50.13	50.13		0	50.13	0	50.13	Devon Sullivan	Loyal	Referral	Running Belt	Sports	0	Gross Sale
15 00:00:00	1011	165	18	1	54.69	54.69		10	49.221	0	49.221	Reese Okafor	Champion	Social	LED Lamp	Home	5.469	Max Sale
15 00:00:00	1012	343	26	3	29.61	88.83		10	79.947	0	79.947	Reese Costa	Champion	Direct	Shampoo Bar	Beauty	8.883	Min Sale
15 00:00:00	1013	379	31	2	37.72	75.44		10	67.896	0	67.896	Mateo Patel	Loyal	Social	Resistance Bands	Sports	7.544	Net Sales
15 00:00:00	1014	395	12	1	110.84	110.84		10	99.756	0	99.756	Noel Novak	Loyal	Organic	Bluetooth Speaker	Electronics	11.084	Net Sales Target
15 00:00:00	1015	447	19	1	11.13	11.13	20	8.904	0	8.904	Morgan Nguyen	New	Social	Cutting Board	Home	2.226	Order Target	
15 00:00:00	1016	470	13	1	60.48	60.48	0	60.48	0	60.48	Jamie Costa	Champion	Paid Search	Webcam	Electronics	0	order_date	
15 00:00:00	1017	568	12	3	110.84	332.52		0	332.52	0	332.52	Hugo Flynn	Loyal	Email	Bluetooth Speaker	Electronics	0	product_id
15 00:00:00	1018	593	36	1	44.06	44.06		0	44.06	0	44.06	Reese Carter	Regular	Paid Search	Puzzle 500pc	Toys	0	Profit Target
15 00:00:00	1019	612	13	1	60.48	60.48		0	60.48	0	60.48	Drew Kowalski	New	Email	Webcam	Electronics	0	order_id
15 00:00:00	1020	644	2	1	24.05	24.05		0	24.05	0	24.05	Tessa Osei	Loyal	Organic	Hoodie	Apparel	0	order_id
15 00:00:00	1021	692	23	1	20.25	20.25		0	20.25	0	20.25	Iris Bauer	Regular	Social	Vitamin C Serum	Beauty	0	product_id
15 00:00:00	1022	772	25	1	5.03	5.03	25	3.7725	1	0	Maya Fischer	Loyal	Email	Sheet Mask	Beauty	5.03	Profit	
15 00:00:00	1023	801	10	1	29.62	29.62		10	26.658	0	26.658	Nora Yamada	Loyal	Organic	USB-C Cable	Electronics	2.962	Profit Target
15 00:00:00	1024	823	34	1	50.13	50.13		0	50.13	0	50.13	Maya Park	Regular	Paid Search	Running Belt	Sports	0	Quantity
15 00:00:00	1025	852	15	3	8.53	25.59		0	25.59	0	25.59	Sasha Diaz	Regular	Paid Search	Smart Plug	Electronics	0	Quantity Target
15 00:00:00	1026	887	30	1	45.18	45.18	5	42.921	0	42.921	Mateo Novak	Champion	Referral	Water Bottle	Sports	2.259	returned	
15 00:00:00	1027	905	13	1	60.48	60.48		0	60.48	0	60.48	Tessa Ruiz	Loyal	Email	Webcam	Electronics	0	

Create a Target as 115% of Current Order

Using $\text{Order Target} = \text{Total Order} * 1.15$

2

Order Target = [Total_order]*1.18

Formatting

Properties

Calculations

order_id

order_id

customer_id

product

quantity

unit_price

Gross Sale

discount_pct

Revenue

returned

Net Sales

Customer

Segment

Channel

Product

Category

Profit

1-08-2025 00:00:00

1001

20

28

1

23.27

23.27

5

22.1065

0

22.1065

Hugo Brooks

Loyal

Paid Search

SPF 50 Lotion

Beauty

1.1635

1-08-2025 00:00:00

1002

41

5

2

58.34

116.68

0

116.68

0

116.68

Owen Nguyen

Regular

Organic

Socks 3-Pack

Apparel

0

1-08-2025 00:00:00

1003

86

31

1

37.72

37.72

25

28.29

0

28.29

Lila Mendez

Regular

Direct

Resistance Bands

Sports

9.43

1-08-2025 00:00:00

1004

97

5

2

58.34

116.68

10

105.012

0

105.012

Jamie Romano

Loyal

Organic

Socks 3-Pack

Apparel

11.668

1-08-2025 00:00:00

1005

108

32

1

51.36

51.36

10

46.224

0

46.224

Tessa Okafor

Loyal

Referral

Foam Roller

Sports

5.136

1-08-2025 00:00:00

1006

113

29

1

31.47

31.47

0

31.47

0

31.47

Cole Okafor

Champion

Referral

Yoga Mat

Sports

0

1-08-2025 00:00:00

1007

115

1

3

67.83

203.49

10

183.141

0

183.141

Owen Petrov

Champion

Paid Search

Crew Tee

Apparel

20.349

1-08-2025 00:00:00

1008

119

22

1

33.84

33.84

0

33.84

0

33.84

Casey Costa

New

Organic

Scented Candle

Home

0

1-08-2025 00:00:00

1009

138

40

1

21.25

21.25

0

21.25

0

21.25

Priya Sullivan

Regular

Email

Art Kit

Toys

0

1-08-2025 00:00:00

1010

141

34

1

50.13

50.13

0

50.13

0

50.13

Devon Sullivan

Loyal

Referral

Running Belt

Sports

0

1-08-2025 00:00:00

1011

165

18

1

54.69

54.69

10

49.221

0

49.221

Reese Okafor

Champion

Social

LED Lamp

Home

5.469

1-08-2025 00:00:00

1012

343

26

3

29.61

88.83

10

79.947

0

79.947

Reese Costa

Champion

Direct

Shampoo Bar

Beauty

8.883

1-08-2025 00:00:00

1013

379

31

2

37.72

75.44

10

67.896

0

67.896

Mateo Patel

Loyal

Social

Resistance Bands

Sports

7.544

1-08-2025 00:00:00

1014

395

12

1

110.84

110.84

10

99.756

0

99.756

Noel Novak

Loyal

Organic

Bluetooth Speaker

Electronics

11.084

1-08-2025 00:00:00

1015

447

19

1

11.13

11.13

20

8.904

0

8.904

Morgan Nguyen

New

Social

Cutting Board

Home

2.226

1-08-2025 00:00:00

1016

470

13

1

60.48

60.48

0

60.48

0

60.48

Jamie Costa

Champion

Paid Search

Webcam

Electronics

0

1-08-2025 00:00:00

1017

568

12

3

110.84

332.52

0

332.52

0

332.52

Hugo Flynn

Loyal

Email

Bluetooth Speaker

Electronics

0

1-08-2025 00:00:00

1018

593

36

1

44.06

44.06

0

44.06

0

44.06

Reese Carter

Regular

Paid Search

Puzzle 500pc

Toys

0

Q Search

Cleaned Order Data

Customer

Segment

Avg Sale

Category

customer_id

discount_pct

Gross Sale

Max Sale

Min Sale

Net Sales

New Sales Target

Order Target

order_date

DAX Functions – Observations

1. **SUM()** was used to calculate Total Sales, Total Profit, and Total Quantity.
2. **DISTINCTCOUNT()** was used to calculate the total of order id number of unique orders.
3. **AVERAGE()** was used to calculate the average sales amount.
4. **TOTALYTD()** was used to calculate Year-to-Date (YTD) Sales using the Date Table.
5. Using DAX reduced manual calculations and improved the accuracy and flexibility of the dashboard.
6. Created a target as 120% of current sales, 125% of Current Quantity, 130% of Current Profit and 115% of current Order.
7. The combination of the Date Table and DAX functions enabled effective time-based analysis, including monthly trends and cumulative sales.

Total net sale:

Home tableCleaned Order Data\$ - %Auto

Structure

Formatting

Properties

New Quick measure measure Calculations

1

Total Net Sale = SUM('Cleaned Order Data'[Net Sales])

order_date	order_id	customer_id	product_id	quantity	unit_price	Gross Sale	discount_pct	Revenue	returned	Net Sales	Customer	Segment	Channel	Product	Category	Profit
02-08-2025 00:00:00	1001	20	\$28	1	23.27	23.27	5	22.1065	0	22.1065	Hugo Brooks	Loyal	Paid Search	SPF 50 Lotion	Beauty	1.1635
02-08-2025 00:00:00	1002	41	\$5	2	58.34	116.68	0	116.68	0	116.68	Owen Nguyen	Regular	Organic	Socks 3-Pack	Apparel	0
02-08-2025 00:00:00	1003	86	\$31	1	37.72	37.72	25	28.29	0	28.29	Lia Mendez	Regular	Direct	Resistance Bands	Sports	9.43
02-08-2025 00:00:00	1004	97	\$5	2	58.34	116.68	10	105.012	0	105.012	Jamie Romano	Loyal	Organic	Socks 3-Pack	Apparel	11.668
02-08-2025 00:00:00	1005	108	\$32	1	51.36	51.36	10	46.224	0	46.224	Tessa Okafor	Loyal	Referral	Foam Roller	Sports	5.136
02-08-2025 00:00:00	1006	113	\$29	1	31.47	31.47	0	31.47	0	31.47	Cole Okafor	Champion	Referral	Yoga Mat	Sports	0
02-08-2025 00:00:00	1007	115	\$1	3	67.83	203.49	10	183.141	0	183.141	Owen Petrov	Champion	Paid Search	Crew Tee	Apparel	20.349
02-08-2025 00:00:00	1008	119	\$22	1	33.84	33.84	0	33.84	0	33.84	Casey Costa	New	Organic	Scented Candle	Home	0
02-08-2025 00:00:00	1009	138	\$40	1	21.25	21.25	0	21.25	0	21.25	Priya Sullivan	Regular	Email	Art Kit	Toys	0
02-08-2025 00:00:00	1010	141	\$34	1	50.13	50.13	0	50.13	0	50.13	Devon Sullivan	Loyal	Referral	Running Belt	Sports	0
02-08-2025 00:00:00	1011	165	\$18	1	54.69	54.69	10	49.221	0	49.221	Reese Okafor	Champion	Social	LED Lamp	Home	5.469
02-08-2025 00:00:00	1012	343	\$26	3	29.61	88.83	10	79.947	0	79.947	Reese Costa	Champion	Direct	Shampoo Bar	Beauty	8.883
02-08-2025 00:00:00	1013	379	\$31	2	37.72	75.44	10	67.896	0	67.896	Mateo Patel	Loyal	Social	Resistance Bands	Sports	7.544
02-08-2025 00:00:00	1014	395	\$12	1	110.84	110.84	10	99.756	0	99.756	Noel Novak	Loyal	Organic	Bluetooth Speaker	Electronics	11.084
02-08-2025 00:00:00	1015	447	\$19	1	11.13	11.13	20	8.904	0	8.904	Morgan Nguyen	New	Social	Cutting Board	Home	2.226
02-08-2025 00:00:00	1016	470	\$13	1	60.48	60.48	0	60.48	0	60.48	Jamie Costa	Champion	Paid Search	Webcam	Electronics	0
02-08-2025 00:00:00	1017	568	\$12	3	110.84	332.52	0	332.52	0	332.52	Hugo Flynn	Loyal	Email	Bluetooth Speaker	Electronics	0
02-08-2025 00:00:00	1018	593	\$36	1	44.06	44.06	0	44.06	0	44.06	Reese Carter	Regular	Paid Search	Puzzle 500pc	Toys	0
02-08-2025 00:00:00	1019	612	\$13	1	60.48	60.48	0	60.48	0	60.48	Drew Kowalski	New	Email	Webcam	Electronics	0
02-08-2025 00:00:00	1020	644	\$2	1	24.05	24.05	0	24.05	0	24.05	Tessa Owe	Loyal	Organic	Hoodie	Apparel	0
02-08-2025 00:00:00	1021	692	\$23	1	20.25	20.25	0	20.25	0	20.25	Iris Bauer	Regular	Social	Vitamin C Serum	Beauty	0
02-08-2025 00:00:00	1022	772	\$25	1	5.03	5.03	25	3.7725	1	0	Maya Fischer	Loyal	Email	Sheet Mask	Beauty	5.03
02-08-2025 00:00:00	1023	801	\$10	1	26.68	26.68	10	26.658	0	26.658	Nora Yamada	Loyal	Organic	USB-C Cable	Electronics	2.962
02-08-2025 00:00:00	1024	823	\$34	1	50.13	50.13	0	50.13	0	50.13	Maya Park	Regular	Paid Search	Running Belt	Sports	0
02-08-2025 00:00:00	1025	852	\$15	3	8.53	25.59	0	25.59	0	25.59	Sasha Diaz	Regular	Paid Search	Smart Plug	Electronics	0
02-08-2025 00:00:00	1026	887	\$30	1	45.18	45.18	5	42.921	0	42.921	Mateo Novak	Champion	Referral	Water Bottle	Sports	2.259
02-08-2025 00:00:00	1027	905	\$13	1	60.48	60.48	0	60.48	0	60.48	Tessa Reid	Loyal	Email	Webcam	Electronics	0
02-08-2025 00:00:00	1028	906	\$30	1	45.18	45.18	5	42.921	0	42.921	Noel Flynn	Champion	Social	Water Bottle	Sports	2.259
02-08-2025 00:00:00	1029	942	\$10	1	29.62	29.62	0	29.62	0	29.62	Naya Kowalski	Loyal	Organic	USB-C Cable	Electronics	0
02-08-2025 00:00:00	1030	970	\$38	1	15.37	15.37	0	15.37	0	15.37	Ezra Sullivan	Regular	Organic	Board Game	Toys	0
02-08-2025 00:00:00	1031	1035	\$10	1	26.68	26.68	10	26.658	0	26.658	Priya Yamada	Loyal	Social	USB-C Cable	Electronics	2.962
02-08-2025 00:00:00	1032	1041	\$16	1	10.61	10.61	0	10.61	0	10.61	Hugo Reyes	Loyal	Social	Throw Pillow	Home	0
02-08-2025 00:00:00	1033	1115	\$16	1	60.48	60.48	0	60.48	0	60.48	Jamie Costa	Champion	Paid Search	Webcam	Electronics	0

Search

Cleaned Order Data

Customer

Segment

Category

Channel

Customer_id

discount_pct

Gross Sale

Net Sales

order_date

order_id

Product

product_id

Profit

quantity

returned

Revenue

Total Net Sale

Date table

Date

day

Month

Quarter

year

Orders table

Product Table

Average sale:

Home tableCleaned Order Data\$ - %Auto

Structure

Formatting

Properties

New Quick measure measure Calculations

1

Avg Sale = AVERAGE('Cleaned Order Data'[Net Sales])

order_date	order_id	customer_id	product_id	quantity	unit_price	Gross Sale	discount_pct	Revenue	returned	Net Sales	Customer	Segment	Channel	Product	Category	Profit
02-08-2025 00:00:00	1001	20	\$28	1	23.27	23.27	5	22.1065	0	22.1065	Hugo Brooks	Loyal	Paid Search	SPF 50 Lotion	Beauty	1.1635
02-08-2025 00:00:00	1002	41	\$5	2	58.34	116.68	0	116.68	0	116.68	Owen Nguyen	Regular	Organic	Socks 3-Pack	Apparel	0
02-08-2025 00:00:00	1003	86	\$31	1	37.72	37.72	25	28.29	0	28.29	Lia Mendez	Regular	Direct	Resistance Bands	Sports	9.43
02-08-2025 00:00:00	1004	97	\$5	2	58.34	116.68	10	105.012	0	105.012	Jamie Romano	Loyal	Organic	Socks 3-Pack	Apparel	11.668
02-08-2025 00:00:00	1005	108	\$32	1	51.36	51.36	10	46.224	0	46.224	Tessa Okafor	Loyal	Referral	Foam Roller	Sports	5.136
02-08-2025 00:00:00	1006	113	\$29	1	31.47	31.47	0	31.47	0	31.47	Cole Okafor	Champion	Referral	Yoga Mat	Sports	0
02-08-2025 00:00:00	1007	115	\$1	3	67.83	203.49	10	183.141	0	183.141	Owen Petrov	Champion	Paid Search	Crew Tee	Apparel	20.349
02-08-2025 00:00:00	1008	119	\$22	1	33.84	33.84	0	33.84	0	33.84	Casey Costa	New	Organic	Scented Candle	Home	0
02-08-2025 00:00:00	1009	138	\$40	1	21.25	21.25	0	21.25	0	21.25	Priya Sullivan	Regular	Email	Art Kit	Toys	0
02-08-2025 00:00:00	1010	141	\$34	1	50.13	50.13	0	50.13	0	50.13	Devon Sullivan	Loyal	Referral	Running Belt	Sports	0
02-08-2025 00:00:00	1011	165	\$18	1	54.69	54.69	10	49.221	0	49.221	Reese Okafor	Champion	Social	LED Lamp	Home	5.469
02-08-2025 00:00:00	1012	343	\$26	3	29.61	88.83	10	79.947	0	79.947	Reese Costa	Champion	Direct	Shampoo Bar	Beauty	8.883
02-08-2025 00:00:00	1013	379	\$31	2	37.72	75.44	10	67.896	0	67.896	Mateo Patel	Loyal	Social	Resistance Bands	Sports	7.544
02-08-2025 00:00:00	1014	395	\$12	1	110.84	110.84	10	99.756	0	99.756	Noel Novak	Loyal	Organic	Bluetooth Speaker	Electronics	11.084
02-08-2025 00:00:00	1015	447	\$19	1	11.13	11.13	20	8.904	0	8.904	Morgan Nguyen	New	Social	Cutting Board	Home	2.226
02-08-2025 00:00:00	1016	470	\$13	1	60.48	60.48	0	60.48	0	60.48	Jamie Costa	Champion	Paid Search	Webcam	Electronics	0

Search

Cleaned Order Data

Customer

Segment

Category

Avg Sale

Channel

Customer_id

discount_pct

Gross Sale

Net Sales

order_date

order_id

Product

product_id

Profit

quantity

returned

Revenue

Orders table

Product Table

MIN and MAX sale

Order	Order Id	Customer Id	Product Id	Quantity	Unit Price	Gross Sale	Discount Pct	Revenue	Returned	Net Sales	Customer	Segment	Channel	Product	Category	Profit
02-08-2025 00:00:00	1001	20	\$28	1	23.27	23.27	5	22.1065	0	22.1065	Hugo Brooks	Loyal	Direct	SFF 50 Loton	Beauty	1.935
02-08-2025 00:00:00	1002	41	\$5	2	116.68	116.68	0	116.68	0	116.68	Owen Nguyen	Regular	Referral	Socks 3-Pack	Apparel	1.632
02-08-2025 00:00:00	1003	86	\$31	2	37.72	37.72	25	28.29	0	28.29	Lila Mendes	Regular	Referral	Resistance Bands	Sports	9.42
02-08-2025 00:00:00	1004	97	\$5	2	58.34	116.68	10	105.012	0	105.012	Jamia Romano	Loyal	Organic	Socks 3-Pack	Apparel	11.668
02-08-2025 00:00:00	1005	108	\$32	1	51.36	51.36	10	46.224	0	46.224	Tessia Okfor	Loyal	Referral	Foam Roller	Sports	5.136
02-08-2025 00:00:00	1006	113	\$29	1	31.47	31.47	0	31.47	0	31.47	Cole Okfor	Champion	Referral	Yoga Mat	Sports	0
02-08-2025 00:00:00	1007	115	\$1	3	67.83	203.49	0	183.141	0	183.141	Owen Petrov	Champion	Referral	Crew Tee	Apparel	20.349
02-08-2025 00:00:00	1008	119	\$20	1	33.84	33.84	0	33.84	0	33.84	Cassy Costa	New	Organic	Scented Candle	Home	0
02-08-2025 00:00:00	1009	128	\$40	1	21.25	21.25	0	21.25	0	21.25	Priva Sullivan	Loyal	Referral	Running Belt	Sports	0
02-08-2025 00:00:00	1010	141	\$34	1	50.13	50.13	0	50.13	0	50.13	Devon Sullivan	Loyal	Referral	Running Belt	Sports	0
02-08-2025 00:00:00	1011	165	\$18	1	54.69	54.69	10	49.221	0	49.221	Reese Okfor	Champion	Social	LED Lamp	Home	5.469
02-08-2025 00:00:00	1012	343	\$26	3	26.61	88.83	10	79.947	0	79.947	Reese Costa	Champion	Direct	Shampoo Bar	Beauty	8.883
02-08-2025 00:00:00	1013	379	\$31	2	37.72	75.44	10	67.896	0	67.896	Mateo Patel	Loyal	Social	Resistance Bands	Sports	7.544
02-08-2025 00:00:00	1014	395	\$12	1	110.84	110.84	0	99.756	0	99.756	Milo Novak	Loyal	Organic	Bluetooth Speaker	Electronics	11.084
02-08-2025 00:00:00	1015	447	\$19	1	11.13	11.13	20	8.904	0	8.904	Morgan Nguyen	New	Social	Cutting Board	Home	2.226
02-08-2025 00:00:00	1016	470	\$10	1	60.48	60.48	0	60.48	0	60.48	Jamia Costa	Champion	Referral	Resistance Bands	Sports	0
02-08-2025 00:00:00	1017	478	\$15	3	430.84	535.23	0	535.23	0	535.23	Morgan Nguyen	Loyal	Referral	Resistance Bands	Sports	0

Sales Data																
1 Min sale = MIN("Cleaned Order Date"/Net Sales)																
order_date	order_id	customer_id	product_id	quantity	unit_price	gross_sale	discount_pct	revenue	returned	net_sales	customer	segment	channel	product	category	profit
02-08-2025 00:00:00	1000	20	\$18	2	23.27	0	0	23.27	0	23.27	Wen Brooks	Regular	Paid Search	Resistance Bands	Apparel	1.6525
02-08-2025 00:00:00	1002	41	\$5	2	58.34	116.68	0	116.68	0	116.68	Owen Nguyen	Regular	Direct	Socks 3-Pack	Apparel	0
02-08-2025 00:00:00	1003	86	\$31	1	37.72	37.72	25	28.29	0	28.29	Lina Mendez	Regular	Direct	Resistance Bands	Sports	9.43
02-08-2025 00:00:00	1004	97	\$5	2	58.34	116.68	10	105.012	0	105.012	Lina Romano	Loyal	Organic	Socks 3-Pack	Apparel	11.668
02-08-2025 00:00:00	1005	108	\$32	1	51.36	51.36	10	46.224	0	46.224	Tessa Okafor	Loyal	Referral	Foam Roller	Sports	5.136
02-08-2025 00:00:00	1006	113	\$29	1	31.87	31.87	0	31.87	0	31.87	Cole Okafor	Champion	Referral	Yoga Mat	Sports	0
02-08-2025 00:00:00	1007	115	\$11	3	67.89	203.67	10	188.141	0	188.141	Owen Patel	Regular	Paid Search	Resistance Bands	Apparel	20.369
02-08-2025 00:00:00	1008	119	\$22	1	33.84	33.84	0	33.84	0	33.84	Cathy Costa	New	Organic	Scented Candle	Home	0
02-08-2025 00:00:00	1009	138	\$40	1	21.25	21.25	0	21.25	0	21.25	Priya Sullivan	Regular	Email	Art Kit	Toys	0
02-08-2025 00:00:00	1010	141	\$34	1	50.13	50.13	0	50.13	0	50.13	Devon Sullivan	Loyal	Referral	Running Belt	Sports	0
02-08-2025 00:00:00	1011	165	\$18	1	54.69	54.69	10	49.221	0	49.221	Reese Okafor	Champion	Social	LED Lamp	Home	5.469
02-08-2025 00:00:00	1012	343	\$26	3	29.61	88.83	10	79.847	0	79.847	Reese Costa	Champion	Direct	Shampoo Bar	Beauty	8.883
02-08-2025 00:00:00	1013	379	\$21	2	54.46	108.92	10	97.096	0	97.096	Matos Patel	Social	Organic	Resistance Bands	Sports	7.544
02-08-2025 00:00:00	1014	395	\$12	1	110.84	110.84	10	99.756	0	99.756	Noel Nwak	Loyal	Organic	Bluetooth Speaker	Electronics	11.084
02-08-2025 00:00:00	1015	447	\$19	1	11.13	11.13	20	8.904	0	8.904	Morgan Nguyen	New	Social	Cutting Board	Home	2.226
02-08-2025 00:00:00	1016	470	\$13	1	60.48	60.48	0	60.48	0	60.48	Jamie Costa	Champion	Paid Search	Webcam	Electronics	0
02-08-2025 00:00:00	1017	568	\$12	3	10.84	32.52	0	32.52	0	32.52	Hugs Flynn	Loyal	Bluetooth	Bluetooth Speaker	Electronics	0
02-08-2025 00:00:00	1018	597	\$48	1	44.06	44.06	0	44.06	0	44.06	Renee Carter	Regular	Paid Search	Resistance Bands	Sports	0
02-08-2025 00:00:00	1019	612	\$13	1	60.48	60.48	0	60.48	0	60.48	Drew Kowalski	New	Email	Webcam	Electronics	0
02-08-2025 00:00:00	1020	644	\$2	1	24.05	24.05	0	24.05	0	24.05	Tessa Osei	Loyal	Organic	Hoodie	Apparel	0

TotalQuantity

	Measure	Measure Column Name	Date	Table
Structure	Relationships	Calculations	Calendars	
Order Date	Customer Id	Product Id	Quantity	Unit Price
Gross Sale	Discount Pct	Revenue	Returned	Net Sales
Customer	Segment	Channel	Product	Category
Profit				

Search

- Cleaned Order Data
- Customer
- Segment
- Avg Sale
- Category
- Channel
- Customer_id
- discount_pct
- Gross Sale
- Max Sale
- Mix sale
- Net Sales
- order_date
- order_id
- Product
- product_id
- Profit
- quantity
- returned
- Revenue
- Total
- Total Net Sale
- Total order
- Total Quantity

Total number of order id

Structure	Relationships	measure	column	table	table
	Relationships	Calculations			Calendars
		1 Total order = DISTINCTCOUNT('Cleaned Order Data'[order_id])			
order_id	order_id	Customer Id	product Id	quantity	Unit price
Gross Sale	discount_pct	Revenue	returned	Net Sales	Customer
Segment	Channel	Product	Category	Profit	
-08-2025-000000	1001	20	\$28	1	23.27
-08-2025-000000	1002	41	\$5	2	58.34
-08-2025-000000	1003	86	\$31	1	37.72
-08-2025-000000	1004	97	\$5	2	58.34
-08-2025-000000	1005	108	\$22	1	51.36
-08-2025-000000	1006	113	\$29	1	31.47
-08-2025-000000	1007	115	\$1	3	63.84
-08-2025-000000	1008	119	\$22	1	33.84
-08-2025-000000	1009	138	\$40	1	21.25
-08-2025-000000	1010	141	\$34	1	50.13
-08-2025-000000	1011	165	\$18	1	54.69
-08-2025-000000	1012	343	\$26	3	29.61
-08-2025-000000	1013	379	\$31	2	37.72
-08-2025-000000	1014	395	\$12	1	110.84
-08-2025-000000	1015	447	\$19	1	11.13
-08-2025-000000	1016	470	\$13	1	60.48
-08-2025-000000	1017	568	\$17	3	33.52
-08-2025-000000	1018	593	\$26	1	44.06
-08-2025-000000	1019	612	\$13	1	60.48
-08-2025-000000	1020	644	\$2	1	24.05
-08-2025-000000	1021	692	\$23	1	20.25
-08-2025-000000	1022	772	\$25	1	5.03
-08-2025-000000	1023	801	\$10	1	29.62
-08-2025-000000	1024	823	\$34	1	50.13
-08-2025-000000	1025	852	\$15	3	8.53
-08-2025-000000	1026	887	\$30	1	45.18
-08-2025-000000	1027	905	\$13	1	60.48
-08-2025-000000	1028	906	\$30	1	45.18
-08-2025-000000	1029	940	\$10	1	29.62

Total	Profit.,YTD	sale
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Structure

Relationships

Innovative

Innovative Accounts

Source

Calculations

Calendars

1

Total Profit

= SUM([Cleaned Order Data].[Product])

der_date	order_id	customer_id	product_id	quantity	Unit_price	Gross_Sale	discount_pct	Revenue	returned	Net_Sales	Customer	Segment	Channel	Product	Category	Profit	
1-2025-00-00000	1001	20	\$28	1	23.27	23.27		5	22.1065	0	22.1065	Hugo Brooks	Loyal	Paid Search	SPF 50 Lotion	Beauty	1.1635
1-2025-00-00000	1002	41	\$5	2	58.34	116.68		0	116.68	0	116.68	Owen Nguyen	Regular	Organic	Socks 3-Pack	Apparel	0
1-2025-00-00000	1003	86	\$31	1	37.72	37.72	25	28.29	0	28.29	14 Mendez	Regular	Direct	Resistance Bands	Sports	9.43	
1-2025-00-00000	1004	97	\$5	2	58.34	116.68	10	105.012	0	105.012	Jamie Romano	Loyal	Organic	Socks 3-Pack	Apparel	11.668	
1-2025-00-00000	1005	108	\$32	1	51.36	51.36	10	46.224	0	46.224	Tessa Okafor	Loyal	Referral	Foam Roller	Sports	5.136	
1-2025-00-00000	1006	113	\$29	1	31.47	31.47	0	31.47	0	31.47	Clare Okafor	Champion	Referral	Yoga Mat	Sports	0	
1-2025-00-00000	1007	115	\$1	3	67.83	203.49	0	183.141	0	183.141	Owen Petrov	Champion	Paid Search	Crew Tee	Apparel	20.349	
1-2025-00-00000	1008	119	\$28.84	1	33.84	33.84	0	33.84	0	33.84	Prisha Castillo	Champion	Organic	Scented Candle	Home	0	
1-2025-00-00000	1009	128	\$40	1	21.25	21.25	0	21.25	0	21.25	Priya Sullivan	Regular	Email	Art Kit	Toys	0	
1-2025-00-00000	1010	141	\$34	1	50.13	50.13	0	50.13	0	50.13	Devon Sullivan	Loyal	Referral	Running Belt	Sports	0	
1-2025-00-00000	1011	165	\$18	1	54.69	54.69	10	49.221	0	49.221	Reese Okafor	Champion	Social	LED Lamp	Home	5.469	
1-2025-00-00000	1012	343	\$26	3	29.61	88.83	10	79.947	0	79.947	Reese Costa	Champion	Direct	Shampoo Bar	Sports	8.883	
1-2025-00-00000	1013	379	\$31	2	37.72	75.44	10	67.696	0	67.696	Mateo Patel	Loyal	Social	Resistance Bands	Beauty	7.544	
1-2025-00-00000	1014	395	\$12	1	110.84	110.84	10	99.756	0	99.756	Noel Novak	Loyal	Organic	Bluetooth Speaker	Electronics	11.084	
1-2025-00-00000	1015	447	\$19	1	11.13	11.13	20	8.904	0	8.904	Morgan Nguyen	New	Social	Cutting Board	Home	2.226	
1-2025-00-00000	1016	470	\$13	1	60.48	60.48	0	60.48	0	60.48	James Costa	Champion	Paid Search	Webcam	Electronics	0	
1-2025-00-00000	1017	568	\$12	3	116.84	350.52	0	332.52	0	332.52	Hugo Flynn	Loyal	Email	Bluetooth Speaker	Electronics	0	
1-2025-00-00000	1018	593	\$26	1	44.06	44.06	0	44.06	0	44.06	Reese Costa	Regular	Paid Search	Uzule 500pc	Toys	0	
1-2025-00-00000	1019	612	\$13	1	60.48	60.48	0	60.48	0	60.48	Drew Kowalski	New	Email	Webcam	Electronics	0	
1-2025-00-00000	1020	644	\$2	1	24.05	24.05	0	24.05	0	24.05	Tessa Osei	Loyal	Organic	Hoodie	Apparel	0	
1-2025-00-00000	1021	692	\$23	1	20.25	20.25	0	20.25	0	20.25	Iris Bauer	Regular	Social	Vitamin C Serum	Beauty	0	
1-2025-00-00000	1022	772	\$25	1	5.03	5.03	25	3.7725	1	0	Maya Fischer	Loyal	Email	Sheet Mask	Beauty	5.03	
1-2025-00-00000	1023	801	\$1	1	29.62	29.62	10	26.658	0	26.658	Nora Yamada	Loyal	Organic	USB-C Cable	Electronics	2.962	
1-2025-00-00000	1024	823	\$34	1	50.13	50.13	0	50.13	0	50.13	Maya Park	Regular	Paid Search	Running Belt	Sports	0	
1-2025-00-00000	1025	852	\$15	3	8.53	25.59	0	25.59	0	25.59	Sasha Diaz	Regular	Paid Search	Smart Plug	Electronics	0	
1-2025-00-00000	1026	887	\$30	1	45.18	45.18	5	42.921	0	42.921	Maya Nguyen	Champion	Referral	Water Bottle	Sports	2.259	
1-2025-00-00000	1027	905	\$12	1	60.48	60.48	0	60.48	0	60.48	Tessa Reid	Loyal	Email	Webcam	Electronics	0	
1-2025-00-00000	1028	906	\$30	1	45.18	45.18	5	42.921	0	42.921	Noel Flynn	Champion	Social	Water Bottle	Sports	2.259	
1-2025-00-00000	1029	942	\$1	1	29.62	29.62	0	29.62	0	29.62	Maya Kowalski	Loyal	Organic	USB-C Cable	Electronics	0	
1-2025-00-00000	1030	970	\$38	1	15.37	15.37	0	15.37	0	15.37	Ezra Sullivan	Regular	Organic	Board Game	Toys	0	
1-2025-00-00000	1031	1035	\$10	1	29.62	29.62	10	26.658	0	26.658	Priya Yamada	Loyal	Social	USB-C Cable	Electronics	2.962	

Q Search

Cleaned Order Data

Customer

Segment

Avg Sale

Category

Channel

Customer_id

discount_pct

Gross Sale

Max Sale

Min Sale

Net Sales

order_date

order_id

Product

product_id

Profit

quantity

returned

Revenue

Total

Total Net Sale

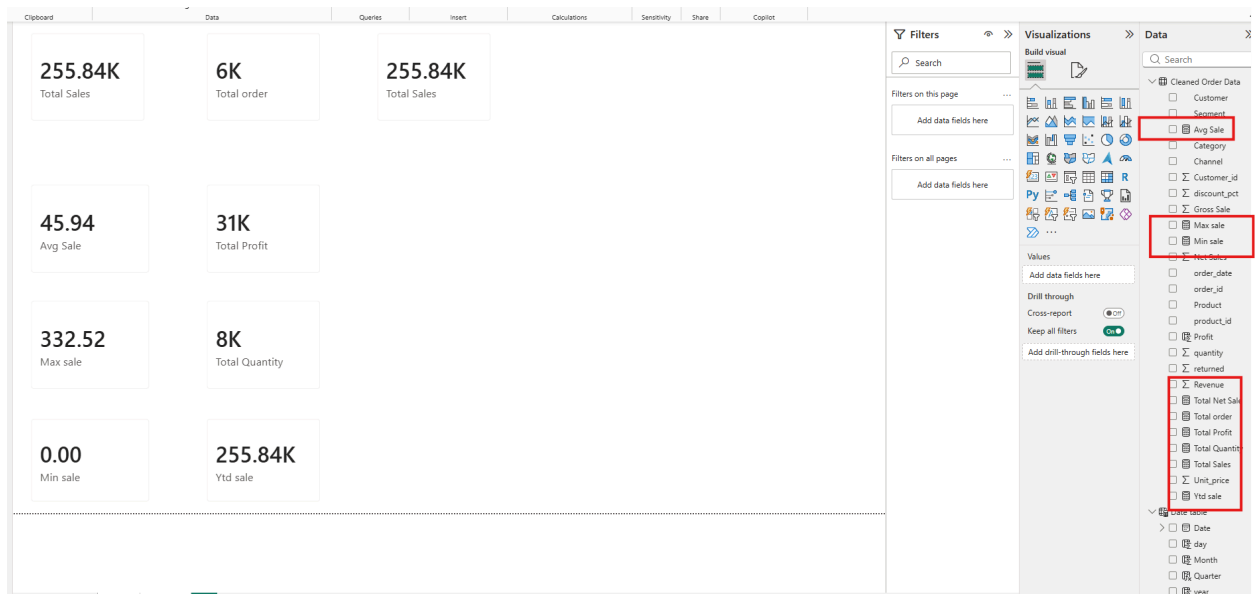
Total order

Total Profit

Total Quantity

Unit Price

Card Displaying



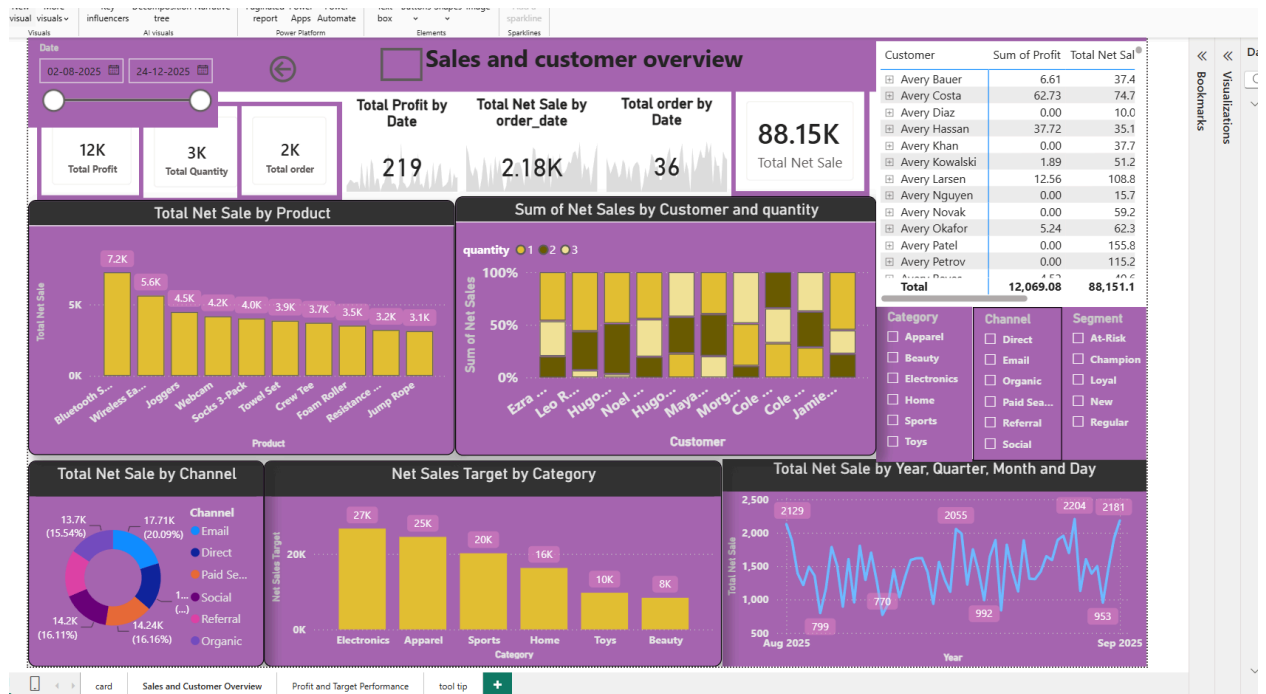
Observations

DAX to create dynamic measures like Total Sales, Total Profit, Profit Margin, YTD, Target Net Sales, Target Quantity, Target Order, and Target Profit. These measures update automatically based on filters and slicers, making the dashboard interactive and helping analyze business performance properly.

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- Multiple Visualizations based on problem statement:
Created 2 dashboards named 1. Sales and Customer Review
2. Target and Performance Analysis.



The Sales and Customer Review Dashboard has

Visualization like

3 KPI's TotalProfit, Total Netsales, Total Order by dates --It Provides an OverallSummary of Business Performance.

Line Chart – Monthly Net Sales Trend –Identifies sales trends over time and highlights peak and low sales months.

Clustered Column Chart – Net Sales by Category –Compares category-wise sales to identify the best- and lowest-performing

Donut Chart – Net Sales by Channel —Shows contributions of each sales channel to total net sales

Bar Chart – Top 10 Customers by Net Sales –Identifies the top customers contributing the highest revenue.

Bar Chart – Top 10 Products by Net Sales –Identifies the top products based on net sales

Matrix –Displays Product and Customer information, Net Sales, Profit and returned status.

Interactive features:

Used a tooltip in a clustered column chart—to compare actual sales and target, compare actual profit and target by category-wise without leaving the chart.

Drill-down: year—Quarter—Month(order date) used in line chart. Users can drill from yearly sales to quarterly and monthly sales.

Slicers: Category, Channel, Segment, and Order Date Slicers are used to filter the dashboard interactively. They help users view data for specific categories, channels, segments, dates, or returned orders.

Bookmarks: Created 4 bookmarks. Reset buttons in each dashboard.

Users can click the Reset button to clear all selected filters and quickly return to the original dashboard state.

Bookmark button: Target Overview & Performance Analysis

Insights of Sales and Customer Review Dashboard

- Identify the highest-selling category.
- Identify the top 10 customers contributing the most sales.
- Identify the top 10 products by Net Sales.
- Compare sales performance across different channels.
- Analyze monthly sales trends.
- Review returned orders and their impact on sales and profit



Target & Performance Analysis

Visualization like

Cards are used: Total Profit Target, Total Sales Target, Total Quantity Target. By date-wise. Provides a quick overview of business targets and overall performance

3 Clustered column charts are used

1.Net Sales vs Sales Target by Category–Compares actual Sales with Sales Target to identify the categories that met or missed their Targets.

2. Profit vs Profit Target by Category–Compares the Actual Profit with Profit Target to identify the profitable status of categories.

3. Quantity vs Quantity by Category–Compares the Quantity with Quantity Target for each category

3 Bar Charts are used:

1.Top Categories by Profit–Identifies the categories generating the highest Profit.

2. Top Segment by Net sales-Identifies the customer segment with the highest sales contribution.

3. Top Channels by Net Sales—Shows which sales channels contribute the highest Net Sales.

Tree Map: Product Profit Contributions—Displays which products contribute the most profit.

Pie chart: Returned and Non-Returned Orders.---Shows the percentage of returned and non-returned orders

Interactive features:

3 Slicers are used: Category, Channel, Segment

Bookmarks used:

Insights of Target & Performance Analysis

- **Compare actual sales, profit, and quantity with their targets.**
- **Identify the most profitable categories.**
- **Find the best-performing sales channel.**
- **Identify the top customer segment.**
- **Analyze product profit contribution.**
- **Monitor the percentage of returned orders.**
- **Help management track target achievement and improve business performance**

9. Insights & Conclusions

- **Key Findings:**
- Sales increased in some months and decreased in others.
- Top customers generated the highest sales.
- Some products and categories performed better than others.
- Some channels and segments contributed more sales.
- A few orders were returned.
- Some categories achieved their sales and profit targets, while others did not.
- **Provide the analysis insights:**

Descriptive

- **Total Sales, Total Profit, Net Sales, and Orders were analysed.**
- **Top Customer, Top Product, Top Channel and Top Segment were Identified.**
- **Monthly sales trend was analyzed.**

Diagnosis

- Top Products and Customers increased Sales.
- Discounts and Returned order was decreased sales.
- Some categories are sold more than others.

Predictive

- **Top Products and Customers are likely to continue generating higher sales.**
- **Categories with low performance may continue to miss target; no action taken.**

Prescriptive

- **Focus on top customers and top Products.**
- **Improve low-performing categories.**
- **Reduce Returned order**
- **Monitor sales, profit, target, and Quantity target regularly.**

10.Conclusions

- **Excel and Power BI were used to clean, analyze, and visualize the sales data.**
- **The dashboards helped identify top customers, top products, top categories, top channels, and top segments.**
- **Sales, Profit, Quantity, and Target performance were compared using interactive visuals.**
- **Features like slicers, drill-down, tooltips, and bookmarks made the report easy to explore.**
- **The dashboard provides useful insights that support better business decisions.**